

Answer Key: Mission 2 Lab Checks

Phase 6: The Mute Button

Question	Answer	Why?
1. To turn off all sound, I should set the volume to...	0	In digital sound, 0 means no power is going to the speaker.
2. If I want to check if the volume is currently silent...	0	<code>if my_volume == 0:</code> checks if the current value is zero.
3. The module I need to import is the...	music	The <code>set_volume</code> tool lives inside the <code>music</code> library.

Phase 7: Volume Control

Question	Answer	Why?
1. Initializing the variable...	volume_level = 128	We start in the middle (half of 255) so the user can go up or down.
2. To make the sound louder...	+ (plus)	Adding to the number increases the intensity of the sound.
3. 255 is the...	maximum (or highest)	The micro:bit cannot process a volume number higher than 255.
4. Update the hardware using...	volume_level	You must pass your "box" (variable) into the <code>set_volume()</code> command.

Phase 8: Creative Design

Question	Answer	Why?
1. The number 0 means the light is...	off (or dark)	0 is no brightness; 9 is full brightness.
2. Which number is missing in the middle?	9	<code>00000:00000:00900:00000:00000</code> creates a single bright dot in the center.
3. Instead of <code>display.show(x)</code> , I will use...	dice_one	You are telling the micro:bit to show your custom Image variable instead of a number.

Teaching Tip for ESL Students

When explaining **Variables** (Phase 7), use the "Box" analogy:

- **The Variable Name** (v) is the label on the outside of the box.
 - **The Value** (128) is the piece of paper inside the box.
 - **The Math** ($v + 16$) is taking the paper out, changing the number, and putting it back.
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